

Program 2: univariate Analysis using Histogram

Aim: To analyze the distribution of a single numerical variable using histogram and KDE plot

Solution:

```
import matplotlib.pyplot as plt
import Seaborn as sns
```

```
Marks = [54,65,34,32,87,21,39,67,43,56, 37,90, 31,34,56,76,98,43]
```

```
sns.histplot(Cdata = Marks, kde = True, bins = 5)
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Density")
```

```
plt.title("WHERE salary")
```

```
plt.show()
```

Program 3: Bivariate Analyze with Regression Line

Title: Analyze Relationship using Scatter Plot and Regression line.

Aim: To visualize the relationships between two variables and fit a regression line

Algorithm:

1. Create Study Hours and Marks data
2. Create a DataFrame.
3. plot study Hours against Marks using regplot()
4. Add labels
5. Observe the relationship

Solution:

```
Import matplotlib.pyplot as plt.
```

```
import Seaborn as sns.
```

```
Hours = [1,2,3,4,5,6,7,8,9]
```

```
Marks = [20,25,30,35,40,45,50,55,60]
```

```
sns.regplot Cx = Hours, y = Marks)
```

```
plt.xlabel("Hours").
```

```
plt.ylabel C" Marks")
```

Alt. Title C" Hours Vs Marks")

plt.show()).

Program 4: Correlation Analysis

Title & Calculating person's correlation Between variables

Aim: To calculate and interpret the correlation between numerical variable

Algorithm: 1. Create a DataFrame using columns, "Study-Hours", "Attendance", "Assignment", "Score", "Marks"

2. Select numerical columns

3. Calculate correlation

4. Display the correlation matrix.

5. Interpret Positive and Negative relationships.

Solution: import matplotlib.pyplot as plt.

import pandas as pd.

data={

"Study-hours" : [2,3,4,5,6,7,8,5,9,10],

```
"Attendance" : [66,70,75,80,85,90,95,78,92,981,
```

```
Assignment" [55,60,65,72,75,80,88,70,90,95],
```

```
"Marks": [45,50,58,65,68,75,82,63,88,84]
```

```
df = pd. DataFrame (data)
```

```
Correlation = df. corr Cnumeric_only = True)
```

```
print C "Pearson correlation")
```

```
print C" Correlation")
```

Program 5:

Title: Analysis of income Distribution using mean, median, Skewness and Histogram with RDE

Aim: To analyze an income dataset using mean, median, and skewness and visualize the income distribution using a histogram with RDECKernal Density Estimation).

Algorithm:

1. Import libraries.

2. Create an income dataframe:

"Income" [15000, 16000, 18000, 20000, 22000, 25000, 28000, 30000, 35000, 100000]

3. Calculate the mean of the income column.

4. Calculate the median of the income column.

5. Calculate the skewness of the income column.

6. Plot the income distribution using a histogram

7. Add a KDE curve to visualize the distribution.

8. Display the graph

9. stop

solution:

```
Import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import pandas as pd
```

```
data = 2
```

```
"Income": [15000, 16000, 18000, 20000, 22000, 25000, 28000, 30000, 35000,100000]
```

```
df= pd. DataFrame (data)
```

```
print C" Mean:", df [" Income"].mean())
```

```
print C" Median:", df [" Income"]. medianc))
```

```
print C" SNewness:", df [" Income"]. SHew()).
```

```
Sns. histplot (data = df, x = "Income", kde = True)
```

```
plt. title (" Income Distribution")
```

```
plt. show()
```

Program 6: Pair plot with regression lines

Title: Multivariate Relationships Exploration using pair plot

Aim: To examine relationships between multiple numerical variables simultan cously using a pair plat.

Algorithm: 1. create a DataFrame using columns "Study Hours", "Attendance

"Assignment score", "Marks",

2. Select numerical columns

3. Display the pairplot matrix using with regression line.

4. Interpret Positive and Negative relationships.

Solution: import matplotlib.pyplot as plt import pandas as pd.

data =

"Study-hours": [2,3,4,5,6,7,8,5,9,10),

"Attendance" : [65,70,75,80,85,90,95,78,92,98],

"Assignment": [65,60,65,72,75,80,88,70,90,95],

"Marks": [45,50,58,65,68,75,82,63,88,94]

df= pd. DataFrame (data)

Sns. pairplot Cdf, Kind = "reg")

plt.show()